

The Sage Who Made SCIENTIFIC COMPUTING TRANSPARENT And AFFORDABLE

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Do you remember Julia, a dynamic programming language for technical computing that we had come across in an

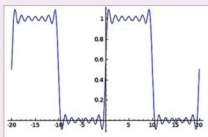


Fig. 1: A communication engineer can make use of Sagemath for signal processing

earlier issue of EFY? Now, let us have a look at SageMath (previously known as SAGE, an abbreviation for System for Algebra and Geometry Experimentation), a cross-platform, python based computer algebraic system for scientific computing. The latest stable release of this software, version 6.9, is bundled in this month's EFY Plus DVD.

Borne out of the frustration of a teacher

William Stein, who was then assistant professor at Harvard University, was unhappy with commercial mathematical software. The frustration was two-fold; one, the huge licence fee charged by commercial giants like MATLAB, Mathematica, Maple, Magma and others, and second, unavailability of the code.

As the code is hidden from the user, mathematicians were not able to cross-check and understand how the computer arrived at such a solution.

When Stein started designing SageMath, he quickly realised that there are a lot of mathematical software for various applications available as open source. Rather than re-inventing the wheel, he

only needed to integrate all of these into a common interface. For this, he chose Python environment. However, the software has since evolved to get thousands of unique lines of code providing new functionalities, apart from just providing a common interface.

Some noteworthy features

SageMath software provides a unique environment for the computation and visualisation of scientific data. Two-dimensional and three-dimensional graphs of symbolic functions and numerical data can be made. Some interesting features are:

User interface. There are three ways in which the user can interact with the software. When the software is running on the server or local system, the user can access it with the help of a Web browser. A text based command line interface using Python has been developed. The user can also access the software as a Python library.

Support for parallel-processing. SageMath is a cross-platform software that allows parallel processing. Users can make use of multiple processors and multi-core processors for parallel processing. They can also use distributed systems to get their computational jobs done.

Use of Python. Instead of writing a new code for every mathematical application, the software reuses other open source software to get the work done. The new code is written only if none of the other software could meet the requirement.

MAIN PACKAGES IN SAGEMATH

Topics	Packages
Algebra	GAP, Maxima, Singular
Algebraic geometry	Singular
Arbitrary precision arithmetic	MPFR, MPFR, MPFI, NTL, mpmath
Arithmetic geometry	PARI/GP, NTL, mwrank, ecmm
Calculus	Maxima, SymPy, GiNaC
Combinatorics	Symmetricon, Sage-Combinat
Linear algebra	ATLAS, BLAS, LAPACK, NumPy, LinBox, iML, GSL
Graph theory	NetworkX
Group theory	GAP
Numerical computation	GSL, SciPy, NumPy, ATLAS
Number theory	PARI/GP, FLINT, NTL
Statistical computing	R, SciPy
Command-line shell	IPython
Database	ZODB, SQLite
Graphical interface	SageMath Notebook, jsMath
Graphics	matplotlib, Tachyon3d, GD, Jmol
Interactive programming language	Python
Networking	Twisted
Differential geometry and tensor calculus	Sage Manifolds